

Energy poverty in the European Union. The art of kaleidoscopic measurement[☆]

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ABSTRACT

Energy poverty measurement in the European Union is a topical issue as existing indicators are disconnected and provide only a partial view of energy-related deprivations. The analysis of unique joint SILC-HBS microdata from Hungary allows for the direct comparison and joint assessment of all headline indicators of energy poverty at the micro level, and yields a series of novel and policy-relevant insights. Importantly, this study finds that existing consensual and expenditure-based measures identify wholly different population segments as energy poor, with less than one third of energy poor households suffering from multiple forms of energy-related deprivations. Different indicators also diverge greatly in terms of incidence, seasonality, cross-country comparability, persistence as well as the socio-demographic background, housing arrangement and living conditions of those identified as energy poor. The presented findings suggest that much of energy poverty may remain hidden using the current data and measurement framework. This calls for new multi-dimensional approaches, the integration of European household surveys, and the exploration of alternative metrics based on the direct measurement of households' energy consumption.

1. Introduction

Energy poverty occurs when households experience inadequate levels of essential energy services at home or in daily life. There is broad consensus that energy poverty in advanced economies is predicated upon a multitude of different vulnerability factors (such as high energy prices, insufficient household incomes, inefficient buildings and appliances, climate, specific household needs), and represents complex, multi-faceted societal challenges (Thomson et al., 2017; Faiella and Lavecchia, 2021; Energy Poverty Advisory Hub, 2022b).

The effects of energy poverty are multiple. There are clear consequences for health and well-being, as extreme indoor temperatures or pollution are linked with a large number of respiratory and cardiovascular diseases, or even premature deaths (Ormandy and Ezratty, 2012; Liddell et al., 2016; Cong et al., 2022). Children, women and the elderly may be at a particular risk (Palmer et al., 2008; Romero

et al., 2018). Energy poverty is also believed to limit one's mobility and transport options, engender sub-standard labour market or economic outcomes, and hold up sustainability, climate action and the green transition (Mattioli et al., 2017; Mikkonen et al., 2020; Adom et al., 2021).

Given its increasing societal costs, energy poverty has become a key priority of policy-makers around the world as of late. This is particularly true of the Global North where energy-related deprivations arise primarily from the unaffordability (rather than limited provision or accessibility) of energy sources and services.¹ Fuelled by the Covid-19 pandemic, the war in Ukraine and ensuing inflation, the issue of energy poverty has resonated particularly among policy-makers of the European Union (EU). Since 2019, the EU has launched several policy

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¹ Note that there is a stark contrast in the perception of energy poverty between different regions. Energy poverty in the Global South is often, and sometimes mistakenly, perceived solely as a problem of limited access to clean fuels, particularly in rural or developing regions. In contrast, in the Global North, it primarily arises from the unaffordability or profitability of modern fuels by households and businesses. While energy poverty in the Global North has been the subject of much independent research, it is usually discussed in the context of multidimensional poverty in the developing world. For further details, see Nussbaumer et al. (2012) and Chan and Delina (2023).

initiatives to address the issue, and obliged Member States to tackle energy poverty in their national strategies.² To streamline these efforts, the European Commission has published two recommendations (EU/2020/1563; EU/2023/2407) to provide guidance for common definitions and measurement, and engaged in extensive policy research coordination in the related fields.

These developments notwithstanding, significant obstacles remain and keep impeding effective policy action at the EU level. At least three important and inter-related dimensions are worth highlighting. First, the conceptual underpinnings of energy poverty, as well as its perceived relationship with more generic forms of poverty are riddled with controversies (Bouzarovski, 2014). The definitional discord is evident when surveying the relevant EU sources: these broadly refer to the inability of a household to access socially and materially necessitated levels of energy services, but differ greatly in terms of the emphasis afforded to key issues surrounding energy poverty debates in the Global North and South — such as the nexus between accessibility, affordability and efficiency concerns, the centrality of thermal comfort against broader deliberations, the accentuation of fairness, cleanness and sustainability considerations, or the integration of standard well-being and capabilities and deprivation perspectives (Day et al., 2016; Chan and Delina, 2023; Al Kez et al., 2024).³ Second, given the multidimensional character of energy poverty and the lack of operational definitions, existing measurement approaches differ greatly in terms of focus, methodology, generosity and character (Della Valle, 2019; Mikkonen et al., 2020; Faiella and Lavecchia, 2021). Indeed, EU monitoring dashboards tend to feature a portfolio of distinct indicators – of both monetary and non-monetary, subjective and objective, qualitative and quantitative, primary and secondary, absolute and relative kind – which makes it particularly hard to make clear assessments and draw straightforward policy conclusions (Heindl, 2015; Thomson et al., 2017; Romero et al., 2018; Cong et al., 2022; Eisfeld and Seebauer, 2022; Energy Poverty Advisory Hub, 2023). Third, available data sources tend to lack integration, cross-country harmonisation or the thematic focus required of sound and consistent measurement. For example, the main European households surveys in use – the EU Statistics on Income and Living Conditions (EU-SILC) and the European Household Budget Survey (EU-HBS) – are based on separate population samples, collected at different times and frequencies, and feature only generic information about households' living conditions and expenditure patterns (Eurostat, 2020, 2022).

At the operational level of national and EU policy-making, measurement has centred around two types of primary indicators (Faiella and Lavecchia, 2021; Energy Poverty Advisory Hub, 2022b; Koukoulfikis and Uihlein, 2022). One of these are subjective or consensual indicators based on households' self-reported energy vulnerability, while the other type includes objective expenditure-based indicators of the affordability of energy services among households. Complementary

indicators (on prices, dwelling characteristics, excess mortality etc.) that are monitored by national governments and the EU's Energy Poverty Advisory Hub (EPAH) typically receive less policy attention due to their incomplete coverage, lack of disaggregation, or indirect relationship with energy outcomes (Energy Poverty Advisory Hub, 2022a; Koukoulfikis and Uihlein, 2022). The preferred focus on consensual and expenditure-based indicators is explicitly stated in the Commission's Recommendation on energy poverty (EU 2020/1563) that lists households' inability to keep home warm, arrears on utility bills, and metrics based on energy-related expenditure patterns as the primary indicators of energy poverty.⁴ Since these indicators are calculated using different European household surveys, no comprehensive analysis of energy poverty is available at the level of individual households in the EU.

The current study aims to address this issue and show the severe limitations of the existing measurement framework for obtaining truthful and reliable information on the extent, concentration and causes of energy poverty across the EU. By exploring a unique sample of joint SILC-HBS microdata from Hungary, this article presents an in-depth analysis of the inter-relationships between main EU indicators of energy poverty for the first time, and produces a series of novel and policy-relevant insights. In particular, it uncovers the statistical relationship between expenditure-based and consensual metrics of energy poverty, and shows the extent to which European households that report inadequate spending on energy may suffer from genuine energy-related deprivations. Its main finding is that different energy poverty measures tend to identify vastly different population segments. Specifically, households that struggle to keep their homes warm or avoid arrears on their utility bills are rarely the ones that spend too little or too much on energy in relative or absolute terms. Different measures also diverge greatly in terms of incidence, seasonality, cross-country comparability, persistence over time, as well as the socio-demographic background and living conditions of those identified as energy poor.

In energy poverty terms, Hungary is broadly representative of the Central and Eastern European (CEE) region, characterised by systemic path-dependencies from the communist period in the form of inherited energy-wasting residential stock and an import-dependent domestic energy demand (Bouzarovski et al., 2016). Available circumstantial evidence and cross-country comparisons suggest, however, that the presented findings also have strong external validity and apply simultaneously in other EU Member States and advanced economies (Devalière and Teissier, 2014; Menyhért, 2022; Palmer et al., 2008). The main policy conclusions are that no single measure of energy poverty can capture all forms and instances of energy-related deprivations, and that a substantial share of such outcomes may remain hidden using only a subset of existing indicators. This calls for improvements in measurement, in particular as far as the international harmonisation of survey data collection and a precise definition of vulnerable households are concerned. The presented findings also call for the exploration of additional new direct measures of energy poverty based on households' energy use, consumption behaviour, dwelling characteristics and minimum energy budgets. Improvements along these lines could greatly enhance both effective social monitoring and targeted policy action.

The remainder of this article is structured as follows. Section 2 introduces the unique Hungarian survey microdata used for the analysis and presents the relevant indicators and measurement aspects. Section 3 compares the energy poverty rate associated with each indicator and assesses the stability of these over time during the calendar year. Section 4 focuses on the overlap between the energy poor populations identified on the basis of each indicator, and demonstrates the considerable disconnect between the various measurable dimensions of energy poverty. Section 5 assesses the role of income and socio-demographic background for energy poverty, discusses the variability of poverty

² The relevant EU policy initiatives include the Clean energy for all Europeans package (2019), the Just Transition Mechanism (2020) or the Fit for 55 package (2021), among others. Implementation at the national level is supported by the National Energy and Climate Plans (NECPs) as well as the EU's long-term budget and the NextGenerationEU financing instrument. For more details, see Menyhért (2023) or the dedicated European Commission website.

³ For example, the dedicated EU website affirms that energy poverty 'occurs when a household must reduce its energy consumption to a degree that negatively impacts the inhabitants' health and wellbeing'. The Commission Recommendation of 2020 (EU/2020/1563) claims that energy poverty 'is a situation in which households are unable to access essential energy services'. The more recent Recommendation (EU/2023/2407), on the other hand, asserts that energy poverty is 'a multidimensional phenomenon driven primarily by three underlying root causes, linked to high energy expenditure in proportion to the household budget, low levels of income, and low energy performance of buildings and appliances'.

⁴ For further details, see the Official Journal of the European Union.

profiles, and demonstrates the importance of dwelling characteristics and housing conditions for energy poverty measurement. Section 6 concentrates on the longitudinal dimension and shows how stability in energy poverty rate over time may obscure significant volatility at the individual and household levels. Section 7 concludes and highlights the relevant policy implications.

2. Data and measurement

As discussed in the introduction, there is no clear conceptualisation or commonly agreed definition of energy poverty in the European Union (Herrero, 2017; Thomson et al., 2017). This allows for a wide range of different measurement strategies among the main potential approaches featured in the key literature (Nussbaumer et al., 2012; Rademaekers et al., 2016; Thomson et al., 2017). These include: (1) expenditure-based metrics based on households' energy-related expenditure (or income) share, (2) consensual metrics based on self-reported assessments of one's housing and energy conditions, (3) direct metrics based on the level of energy services consumed, and (4) outcome-based metrics focusing on energy poverty-relevant outcomes (e.g. disconnections, cold-related mortality), as well as their various combinations and composites.

While a small number of European countries have adopted an official energy poverty definition and developed national indicators of various kinds (Rademaekers et al., 2016; Pye et al., 2017), official statistical measurement and monitoring in the EU is based on a limited set of consensual and expenditure-based indicators. This is due mainly to the lack of accessible, harmonised and cross-country comparable microdata, which puts standard household surveys at the centre of measurement efforts (Pye et al., 2017; Faiella and Lavecchia, 2021; Energy Poverty Advisory Hub, 2022a). In particular, two thematic EU surveys are used. The EU Statistics on Income and Living Conditions (EU-SILC) collects comparable cross-sectional and longitudinal data on households' income, social situation and living conditions at annual frequencies, and is used to derive consensual measures of energy poverty based on households' self-reported energy-related deprivations (Eurostat, 2020). The EU Household Budget Survey (EU-HBS), on the other hand, is a harmonised collection of national budget surveys focusing on households' consumption structure, and is the main source for deriving expenditure-based measures of energy poverty (Eurostat, 2022). Besides the long-standing concerns about timeliness and cross-country comparability, the main limitation of the existing survey infrastructure is the lack of integration: the underlying national HBS and SILC data files are compiled on the basis of different definitions, population samples and survey methods (Donatiello et al., 2014; Menyhért et al., 2021). This makes it generally impossible to jointly examine the consensual and expenditure-based aspects of households' energy situation, and properly understand the forms, causes and consequences of energy-related deprivations.

The empirical analysis in the paper attempts to achieve just this, using a unique set of household survey microdata from Hungary. In energy poverty terms, Hungary is broadly representative of the CEE region as a whole — and particularly the neighbouring EU countries of Slovakia, Czechia and Poland. As elsewhere in the region, systemic path-dependencies have played a key role in the recent emergence of domestic energy deprivation: Hungary inherited an inefficient housing stock and a rather specific supply situation (e.g. strong dependence on Russian gas imports, predominance of district heating systems) specific to socialist economies, as well as the gradual restructuring of the energy sector away from the previous centrally planned system over the last 30 years which led to significant price increases (Bouzarovski et al., 2016).

Specifically, the data used for the current analysis come from joint SILC-HBS national samples from Hungary, originally acquired from the Hungarian Central Statistical Office for the purposes of previous JRC research on absolute poverty measurement (Menyhért et al., 2021).

The Hungarian microdata are unique in that both the SILC and HBS data components are collected based on the same (overlapping) population sample, which allows for the joint analysis of households' living conditions and expenditure patterns at the individual level.⁵ This represents a significant advantage over alternative merged SILC-HBS datasets produced by Eurostat and selected NSIs: these latter are based on assumption-based statistical matching and conditionally random imputations across different sources, and yield reliable estimates only at the level of population groups (rather than individual households) (Donatiello et al., 2014; Balestra and Oehler, 2023; Maia et al., 2023). In addition, the joint Hungarian dataset also features a longitudinal component as well as auxiliary information in a number of survey domains (such as the exact date of the interview), which permits additional empirical analysis compared to standard SILC and HBS distributions offered by Eurostat.

Importantly, the Hungarian microdata allow for the joint analysis of all primary indicators of energy poverty used in the EU — consensual measures of households' self-reported energy situation and expenditure-based measures targeting insufficient or excessive energy spending. The Commission Recommendation (EU/2020/1563) on energy poverty lists the specific metrics in question and defines the particular survey variables and techniques required for their calculation.⁶ In line with the Recommendation, the empirical analysis in this study concentrates on the following five primary indicators of energy poverty:

- the share of population having arrears on utility bills (SILC [variable HS021], henceforth *Arrears*);
- the share of population not able to keep home adequately warm (SILC [HH050], *HomeWarm*);
- the share of population living in households where the energy expenditure-to-income ratio is more than twice the national median (HBS [HE045, HE0722, HH099], *HighShare*);
- the share of population living in households where the absolute level of energy expenditures is less than half the national median (HBS [HE045, HE0722], *LowExpense*)⁸;
- the share of population living in households where the energy expenditure share exceeds 30% relative to total expenditures (HBS [HE045, HE0722, HE00], *FixThreshold*)⁹.

At least four aspects are worth noting in relation to these metrics. First, each of them are derived separately on either the SILC or the HBS data component and therefore do not lend themselves to joint micro-level analysis in most EU countries.¹⁰ Second, they have limited comparability due to differences in scope: while the SILC-based indicators refer exclusively to energy outcomes at home, the

⁵ See the dedicated [website](#) of the Hungarian Statistical Office (HCSO) for more technical and methodological details. Note that the Czech Statistical Office (CZSO) also collects the SILC and HBS surveys using overlapping samples (see its [website](#) for details), but the joint microdata have not yet been made available for research purposes.

⁶ It also lists a series of complementary indicators, referring either to energy prices or more generic indicators of material distress (such as housing deprivation). Rademaekers et al. (2016) and Castaño-Rosa et al. (2019) provide a critical review of the conceptual foundations, genealogy, relative merits and shortcomings of the relevant indicators.

⁷ This indicator is sometimes referred to as 2M in the relevant literature.

⁸ This indicator is referred to as M/2 in the relevant literature.

⁹ The *FixThreshold* metric is not explicitly listed in the relevant Recommendation. It is a revised version of the first formal definition of energy poverty based on an absolute 10% threshold for the energy expenditure share (Boardman, 1991; Romero et al., 2018). The 30% threshold applied for the current analysis is chosen to be in line with Hungarian households' typical energy expenditure patterns and the implied energy poverty rate based on existing indicators (Heindl, 2015; Thomson et al., 2017).

¹⁰ In the joint Hungarian dataset used for the current analysis, only the SILC-based income variables were present and used for the calculation of the (HBS-based) *HighShare* indicator.

HBS-based indicators consider both housing-related and transportation-related energy expenditures.¹¹ Available empirical analyses suggest that expenditure patterns for housing-related and transportation-related energy use are rather different (Simcock et al., 2021; Menyhért, 2022; Vandyck et al., 2023). Third, none of these measures target or capture households' energy behaviour, energy consumption or energy efficiency, and represent a very indirect way to measuring energy poverty (Rademaekers et al., 2016; Thomson et al., 2017; Romero et al., 2018; Cong et al., 2022). Fourth, none of the selected indicators measure the depth of households' energy poverty (i.e. the shortfall relative to the poverty threshold) or respond directly to changes in households' financial situation or purchasing power.¹²

In short, the main advantage of existing measures of energy poverty lies in their simplicity and straightforward measurement based on existing European household surveys. While it is widely believed that improving the existing survey architecture and directly monitoring households' energy consumption and energy efficiency would be a boon to energy poverty measurement, conventional wisdom holds that different measures are broadly comparable and identify largely similar underlying conditions (Al Kez et al., 2024). The analysis in the following sections challenges this view and shows that swapping measurement tools produces important kaleidoscopic changes in the observable patterns of energy poverty.

3. Incidence and seasonality of energy poverty

Using the different measures of energy poverty listed above, let us start by comparing the relevant energy poverty rates as a baseline exercise. While technically feasible using different surveys, this tabulation already exploits the joint feature of the Hungarian data in terms of eliminating common sampling-related differences that characterise income and expenditure surveys across the EU (Eurostat, 2020, 2022; Menyhért et al., 2021).

Fig. 1 shows the level of energy poverty by measure using the full sample ($n = 15540$) of the 2018 survey wave. The corresponding values range between 5.8% for the *HomeWarm* and 17.7% for the *LowExpense* metric. Such variations in energy poverty rates across the different metrics are not unique to Hungary and are comparable to the what may be observed in other EU Member States.¹³ This highlights large differences in perspective across individual indicators and demonstrates the centrality of measurement to meaningful policy discussions of energy poverty. The joint comparison also paints a much fuller picture of energy poverty than any single indicator alone, as it hints at the relative importance of affordability, accessibility or efficiency-related deprivations.¹⁴

While such headcount ratios often feature in annual or multi-annual statistics, it is worth noting that they represent point-in-time estimates

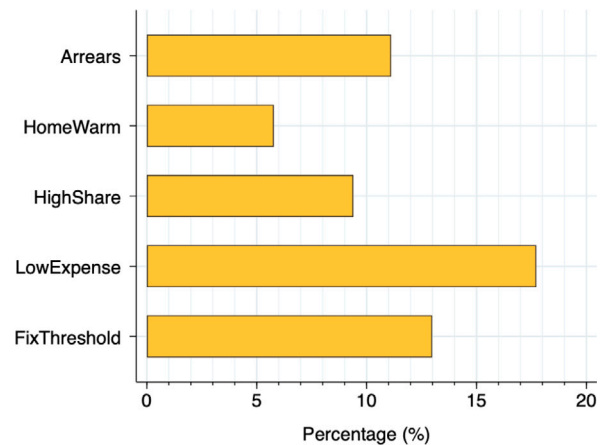


Fig. 1. Energy poverty rates by measure. Notes: The figures refer to 2018, and are based on own analysis of joint SILC-HBS microdata from Hungary. See the detailed explanation of the measures in Section 2.

of energy poverty pertaining to a brief data collection period. Despite ample empirical evidence showing that European households' energy use fluctuates considerably over the calendar year, with much higher consumption in the winter period, this sampling-related aspect of energy poverty measurement remains vastly overlooked (Liddell et al., 2016; Faiella and Lavecchia, 2021).¹⁵ Indeed, one can reasonably expect energy poverty to be subject to seasonal fluctuations, which may further aggravate perception bias, recall errors and potential ambiguity surrounding certain survey-based indicators.¹⁶ Own analysis of EU-SILC microdata from 2019 shows that data collection spans multiple quarters of the year in most EU Member States, and that energy poverty rates are, respectively, 1.2 p.p. and 1.8 p.p. lower in the summer than in winter on average.¹⁷ Moreover, seasonal differences may reach up to 5 percentage points in selected Member States, which raises serious doubts about the cross-country comparability of energy poverty statistics in the EU.¹⁸

While EU household surveys only indicate the quarter of data collection, detailed information on interview dates in the Hungarian microdata makes it possible to effectively monitor trends in energy poverty within the calendar year. Fig. 2 presents the week-by-week evolution of the energy poverty rate associated with each metric during the relevant data collection period of March and April of 2018. It reveals that SILC-based consensual measures of energy poverty are highly volatile, and decrease substantially with the arrival of the warmer season: *Arrears* decreased by 5.1 p.p. between early March and late April (12.9% vs. 7.8%), while *HomeWarm* decreased by 8.4 p.p. during

¹¹ According to the HICP methodological manual compiled by Eurostat, energy expenditures are defined as household spending on both (1) electricity, gas, solid fuels and heat energy [CP045] and (2) fuels and lubricants for personal transport equipment [CP0722]. See Eurostat (2018) for further details on the methodology.

¹² As the analyses by Thema and Vondung (2020) and Menyhért (2022) show, the income elasticity of consensual measures of energy poverty is rather small, while the response of most expenditure-based measures to inflation and changes in households' purchasing power depends on distributional factors and is rather unpredictable.

¹³ Using EU-wide SILC and HBS microdata from 2015, the largest and smallest within-country differences are observed in Greece (32.9 p.p.) and Slovakia (5.5 p.p.), respectively. Besides the absolute differences, the relative ranking of energy poverty indicators in a country also displays large variations across the EU. See some relevant statistics in the Eurostat Data Browser (data codes: ILC_MDES01, ILC_MDES07) for details.

¹⁴ For example, the large number of Hungarian households that spend comparatively little on energy but avoid running arrears or sacrificing thermal comfort appears to suggest large variations in energy efficiency.

¹⁵ Recent Eurostat data shows that Europe's gas consumption is 2.5-times higher in winter than in the summer. Seasonal fluctuations in electricity and oil are smaller, amounting to 30% and 20%, respectively. For more details, see the relevant monthly energy statistics by Eurostat (NRG_CB data series).

¹⁶ Note that the SILC question 'Can your household afford to keep its home adequately warm?' (variable HH050, used for the *HomeWarm* metric) qualifies neither the meaning of 'adequately warm' nor the time period under consideration. The other survey questions related to arrears on utility bills (form the SILC) and energy expenditures (from the HBS) are less ambiguous and clearly specify the reference period in question (i.e. typically the last 12 month).

¹⁷ These figures naturally refer to the *Arrears* and *HomeWarm* metrics as of 2019. Qualitatively similar results are obtained on the basis of other EU-SILC waves, as well.

¹⁸ Similar sampling issues are equally present in relation to expenditure-based measures of energy poverty, as national EU-HBS files are also collected in different survey periods.

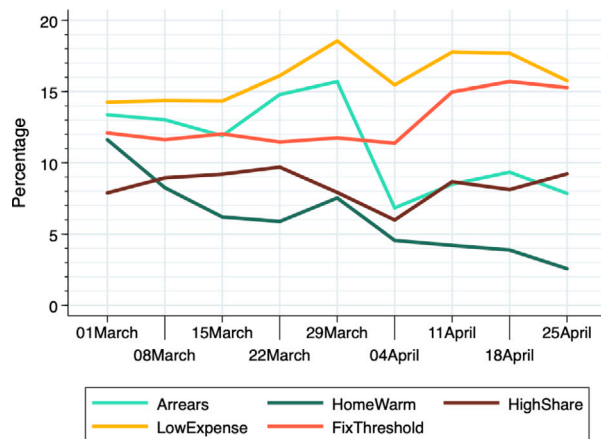


Fig. 2. Seasonality of energy poverty. Notes: The figures refer to weekly aggregates from March and April of 2018, and are based on own analysis of joint SILC-HBS microdata from Hungary. See Section 2 for more explanations.

the same period (11.0% vs. 2.6%).¹⁹ Fig. 2 also shows that expenditure-based metrics display relative stability over time, for a number of possible reasons.²⁰ This suggests that, until further improvements in the cross-country harmonisation of data collection periods, expenditure-based measures of energy poverty may yield more representative, dependable and internationally comparable outcomes than consensual indicators (Rademaekers et al., 2016).

4. Overlap between energy poverty dimensions

The real advantage of the joint Hungarian SILC-HBS microdata is that it allows to explore the statistical relationship and overlap between the different energy poverty measures at the level of individual households. The histogram in Fig. 3 plots the distribution of the national population in 2018 based on their exposure to energy poverty, according to the number of metrics by which a household is considered energy poor. It shows that 55.9% of the population is not energy poor based on either measure, while the strong majority (30.8%) of the affected population is energy poor in a single dimension only. The histogram also reveals that the share of population suffering from multiple (3+) forms of energy poverty simultaneously is very low (2.6%). This suggests that the number and share of households that are energy poor is much higher than what measurement based on a single indicator or data source would suggest. Specifically, it implies that hidden energy poverty, a concept increasingly discussed in the literature (Cong et al., 2022; Eisfeld and Seebauer, 2022), is not a residual but an inherent feature of the current framework of energy poverty measurement. Secondly, Fig. 3 also implies that vulnerable

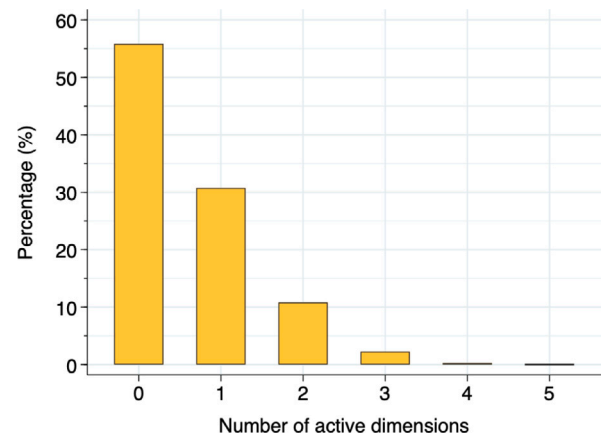


Fig. 3. Concentration of energy poverty dimensions. Notes: The figures refer to 2018, and plot the distribution of households according to the number of active energy poverty dimensions, based on joint SILC-HBS microdata from Hungary.

households tend to self-select into particular forms of energy poverty, and that consensual and expenditure-based measures likely capture different dimensions of energy-related deprivations.

To better understand the inter-relations between different energy poverty dimensions, one may equally consider the degree of overlap in relation to each pairwise metric combination. As expected, the relevant Jaccard indices reveal a very high degree of misclassification between most measurement pairs: the relative size of intersection remains well below 10% in most cases, except for the HBS-based *HighShare* and *FixThreshold* metrics driven by the same underlying spending patterns, and the SILC-based *Arrears* and *HomeWarm* metrics of self-perceived energy deprivation.²¹ This appears to signal that, given the possibly large variations in households' energy needs and energy efficiency, it is very hard to find appropriate indirect indicators that would reliably signal energy-related deprivations.

Fig. 4 offers a graphical illustration of the joint relationship between the various energy poverty measures under consideration. The relevant Venn diagram features area-proportional circles to display the set relationships in question, and confirms the large separation between the respective energy poor populations.²² Specifically, it highlights that struggling to avoid payment arrears or keep home adequately warm is largely unrelated to households' energy expenditure patterns. This is consistent with the conclusions of previous studies from other European countries, and calls for a critical re-assessment of existing energy poverty measures and available survey information (Ormandy and Ezratty, 2012; Curtis and Pentecost, 2015; Faiella and Lavecchia, 2021). In particular, it calls for direct measurement of households' energy practices and consumption to help identify those that are most vulnerable to energy-related deprivations.

5. The role of socio-demographic background and living conditions

The joint Hungarian microdata are also useful to explore the relationship between energy poverty measures and households' income and socio-demographic background in a comparative manner. Identifying

¹⁹ The close correlation of consensual measures with climatic factors gets even more conspicuous when one compares the presented energy poverty trends with the observed historical temperature data collected during the same period. These latter show, for example, that a marked decline in temperatures over the second half of March 2018 were likely responsible for the kinks in the relevant *Arrears* and *HomeWarm* series. For further details, see the relevant site for wunderground.com.

²⁰ Note that, in the Hungarian microdata, expenditure-based measures of energy poverty concern essentially the same reference period as consensual measures (i.e. interview date/diary period). The stability of expenditure-based energy poverty measures over time is likely due to institutional factors (such as combined billing, bill smoothing or flat rate payments for energy services) or structural features of the metrics themselves (such as the distributional aspect of the *HighShare* or *LowExpense* metrics).

²¹ The Jaccard index is a widely used statistic to measure the similarity between sample sets, defined as the size of the intersection relative to the union.

²² Note that, due to visualisation constraints, the position of the respective sets in Fig. 4 is the result of an optimisation procedure and may not accurately reflect all set relationship accurately.

Table 1
Regression results of energy poverty on socio-demographic characteristics.

	Dependent variable				
	Arrears	HomeWarm	HighShare	LowExpense	FixThreshold
Panel A. Household-level characteristics					
Income quintile #1	0.173 (.011)	0.123 (.009)	0.132 (.012)	0.338 (.015)	0.018 (.013)
Income quintile #2	0.049 (.011)	0.024 (.009)	0.025 (.011)	0.104 (.014)	0.004 (.013)
Income quintile #4	−0.024 (.011)	−0.024 (.009)	−0.016 (.011)	−0.089 (.015)	−0.005 (.013)
Income quintile #5	−0.052 (.011)	−0.029 (.009)	−0.034 (.011)	−0.154 (.015)	−0.034 (.013)
Urban area	0.014 (.008)	−0.010 (.007)	−0.028 (.009)	0.147 (.011)	−0.080 (.010)
Rural area	0.019 (.009)	0.012 (.007)	0.027 (.009)	−0.032 (.012)	0.038 (.010)
Small household	−0.039 (.008)	0.027 (.007)	0.024 (.008)	0.181 (.010)	0.034 (.009)
Large household	0.048 (.015)	0.019 (.012)	−0.048 (.015)	−0.070 (.020)	−0.020 (.017)
Constant	0.080 (.010)	0.029 (.009)	0.076 (.011)	0.059 (.014)	0.138 (.012)
R squared	.078	.058	.049	.232	.027
Nr of obs.	6952	6952	6952	6952	6952
Panel B. Individual characteristics					
Child	0.049 (.011)	−0.012 (.008)	−0.023 (.010)	−0.036 (.013)	−0.024 (.012)
Youth	0.026 (.011)	0.012 (.008)	−0.001 (.011)	−0.033 (.014)	−0.012 (.012)
Elderly	−0.061 (.010)	−0.014 (.008)	0.002 (.009)	0.035 (.012)	0.018 (.011)
Female × Child	0.009 (.014)	0.010 (.010)	0.008 (.013)	0.013 (.017)	−0.011 (.015)
Female × Youth	0.023 (.014)	0.003 (.011)	−0.017 (.013)	0.032 (.017)	−0.023 (.016)
Female × Prime-aged	0.001 (.007)	0.000 (.005)	−0.006 (.006)	0.017 (.008)	−0.015 (.007)
Female × Elderly	0.002 (.011)	0.022 (.008)	−0.015 (.011)	0.110 (.014)	−0.005 (.012)
Constant	0.111 (.005)	0.057 (.004)	0.101 (.005)	0.158 (.006)	0.138 (.005)
R squared	.013	.001	.001	.018	.002
Nr of obs.	15 540	15 540	15 540	15 540	15 540

Notes: The presented results are based on sample-weighted OLS regressions featuring different metrics of energy poverty as the dependent variable and categorical variables on individuals' and households' socio-demographic background on the right hand side, using the full-cross section of the 2018 wave of the joint Hungarian SILC-HBS microdata. The reported estimates represent conditional mean differences in the energy poverty rate across different population segments as compared to the reference category – medium-sized households in the 3rd equivalised income quintile living in moderately densely populated areas in Panel A, and prime-aged male individuals in Panel B – captured by the constant term. Standard errors are in parenthesis, with statistical significance denoted by stars at the usual probability levels (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

Table 2
Regression results of energy poverty on dwelling characteristics.

	Dependent variable				
	Arrears	HomeWarm	HighShare	LowExpense	FixThreshold
Detached house	0.025 (.015)	0.024 (.009)	0.011 (.012)	−0.030 (.021)	0.069 (.017)
Apartment (small building)	−0.011 (.022)	−0.020 (.009)	−0.077 (.011)	0.088 (.030)	−0.069 (.025)
Apartment (large building)	−0.022 (.016)	−0.048 (.009)	−0.067 (.011)	0.131 (.022)	−0.060 (.018)
Other accommodation	0.102 (.036)	0.031 (.009)	0.090 (.011)	−0.033 (.050)	−0.017 (.041)
1 or 2 rooms	0.034 (.010)	0.035 (.009)	−0.006 (.009)	0.203 (.014)	−0.006 (.012)
5 or more rooms	−0.027 (.009)	−0.017 (.008)	−0.024 (.009)	−0.090 (.013)	−0.021 (.011)
Owner with mortgage	0.034 (.011)	−0.014 (.007)	−0.052 (.008)	−0.079 (.015)	−0.018 (.013)
Tenant (market rate)	0.047 (.016)	0.005 (.012)	−0.043 (.015)	0.063 (.023)	−0.085 (.019)
Tenant (reduced rate)	0.143 (.019)	0.062 (.012)	−0.071 (.015)	0.238 (.027)	−0.085 (.022)
Free accommodation	0.037 (.015)	0.003 (.012)	−0.031 (.015)	0.143 (.020)	−0.051 (.017)
Constant	0.074 (.015)	0.062 (.012)	0.142 (.015)	0.206 (.020)	0.133 (.017)
R squared	.017	.017	.022	.132	.043
Nr of obs.	6960	6960	6960	6960	6960

Notes: The presented results are based on sample-weighted OLS regressions featuring different metrics of energy poverty as the dependent variable and categorical variables on individuals' and households' dwelling characteristics on the right hand side, using the full cross-section of the 2018 wave of the joint Hungarian SILC-HBS microdata. The reported estimates represent conditional mean differences in the energy poverty rate across different population segments as compared to the reference category – outright owner households living in semi-detached houses with 3 or 4 rooms – captured by the constant term. Standard errors are in parenthesis, with statistical significance denoted by stars at the usual probability levels (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

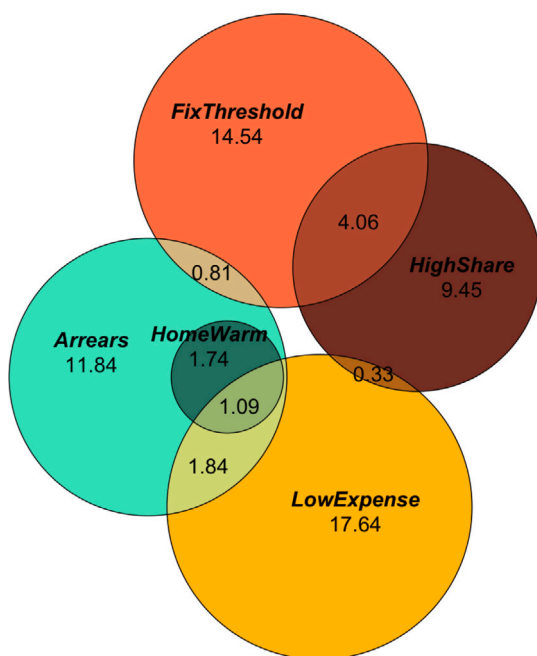


Fig. 4. Overlap between energy poverty dimensions. Notes: Own illustration based on joint Hungarian SILC-HBS microdata from 2018, produced with the R package 'eulerr'. The figures represent the percentage point share of the energy poor by dimension and pairwise combination. Note that, due to visualisation constraints, the position of the respective sets is the result of an optimisation procedure and may not represent all set relationship accurately.

the populations segments with the highest and lowest concentrations of energy poverty by dimension is important from a policy perspective, as low income may not be the only or even principle driver of energy-related deprivations. In fact, the empirical analysis of the Hungarian microdata shows that the statistical association between (equivalised) household income and energy poverty status varies markedly across the different indicators: energy poverty based on the *FixThreshold* metric is virtually unrelated to household income, energy poverty based on the *HighShare* metric concerns mainly very low-income households, while energy poverty based on consensual metrics is concentrated among households with moderately low-income. In line with existing evidence from other countries, this suggests that a significant part of energy poverty persists among non-financially distressed households, and that income support alone is unlikely to eliminate all its forms and manifestations (Thomson et al., 2017; Romero et al., 2018).

Regression analysis can help reveal variations in energy poverty rates across different segments of the population. Table 1 presents a concise summary of the role of socio-demographic characteristics in explaining energy poverty rates associated with different measures at the household and individual levels. Panel A focuses on conditional differences at the household level across income quintiles, settlement types and household of different size. It reveals several important findings. First, the explanatory power of socio-demographic characteristics varies substantially across different measures of energy poverty: the RHS variables explain 23.2% of all variation in the *LowExpense* metric but only 2.7% in the *FixThreshold* metric. Second, while higher-income households experience consistently lower rates of energy poverty, the concentration on income displays marked differences across measures: the gap in the poverty rate between the lowest and highest income quintiles is 49.2 p.p. for the *LowExpense* metric and only 5.2 p.p. for the *FixThreshold* metric. Third, both the size and sign of differences by settlement type and household size changes substantially from one measure to the next: the urban-rural gap in the conditional energy

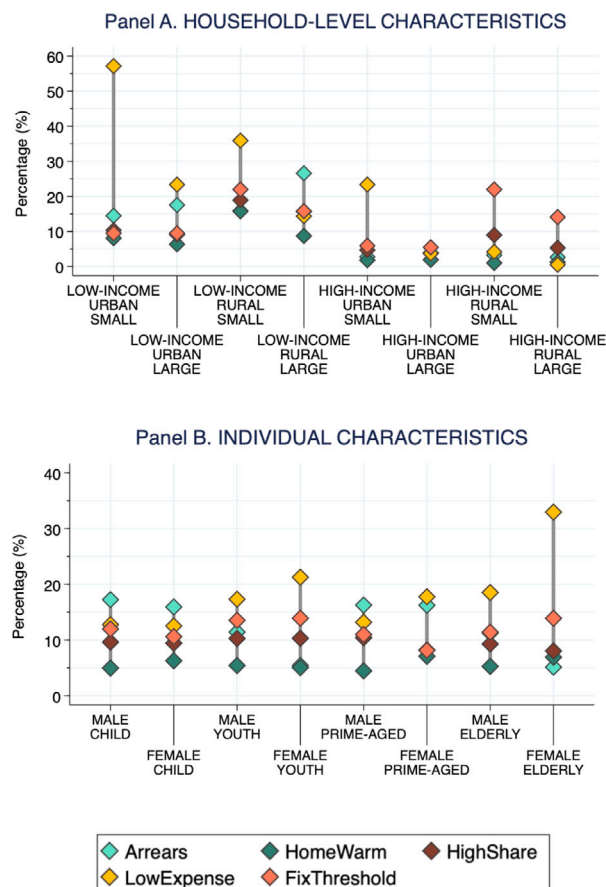


Fig. 5. Energy poverty profiles by measure. Notes: The figures refer to 2018, and illustrate the relationship between energy poverty and socio-demographic characteristics, based on own analysis of joint SILC-HBS microdata from Hungary. Income groups are created on the basis of the national (equivalised) median, household size is split between 1–2 vs. 3+ members (SMALL vs. LARGE), while CHILD/YOUTH/PRIME-AGED/ELDERLY stand for age groups 0–14/15–24/25–64/65+.

poverty rate varies between –11.8 p.p. (the *FixThreshold*) and +17.9 p.p. (*LowExpense*), while the gap between small and large households ranges between –8.7 p.p. (*Arrears*) and +25.1 p.p. (*LowExpense*). Panel B of Table 1 focuses on the individual level, and reveals considerably smaller variations along age and gender lines. The low goodness-of-fit scores and statistically mostly insignificant parameter estimates indicate that horizontal differences in energy poverty are mostly driven by household-level (rather than individual-level) factors, even though some variations do emerge with regard to particular measures.²³

Fig. 5 plots the conditional energy poverty rates for each measure by population segment, and reveals markedly different poverty profiles. Panel A, focusing on household-level characteristics, shows that energy poverty varies substantially across both social groups and indicators in a cross-cutting manner. Different measures tend to yield very different energy poverty rates and rankings, especially when using the *LowExpense* metric or considering small, urban, low-income households.²⁴

²³ For example, these include disproportionately high energy poverty among children based on the *Arrears* metric or among elderly women based on the *LowExpense* metric.

²⁴ Energy poverty varies within an average range of 25.4 p.p. across population groups for a given measure (vs. 56.6 p.p. for *LowExpense*), and within an average range of 20.5 p.p. across measures for a given population group (vs. 49.1 p.p. range for small, urban, low-income households). The average rank correlation is 51% and 42.5% across metrics and population groups, respectively.

Table 3
Energy poverty rates by housing condition.

	Variable of interest				
	Arrears	HomeWarm	HighShare	LowExpense	FixThreshold
Leaking roof or damp walls	0.061 / 0.282	0.032 / 0.146	0.095 / 0.089	0.154 / 0.255	0.137 / 0.107
No bath or shower	0.102 / 0.367	0.050 / 0.267	0.094 / 0.093	0.169 / 0.392	0.131 / 0.102
No indoor flushing toilet	0.102 / 0.341	0.049 / 0.262	0.094 / 0.098	0.168 / 0.389	0.131 / 0.102
Not enough light	0.096 / 0.278	0.050 / 0.139	0.094 / 0.095	0.167 / 0.286	0.132 / 0.102
Noise from street or neighbours	0.100 / 0.222	0.054 / 0.099	0.095 / 0.082	0.171 / 0.240	0.133 / 0.100
Pollution or grime	0.098 / 0.230	0.050 / 0.127	0.091 / 0.119	0.171 / 0.234	0.131 / 0.118
Crime or violence in the area	0.103 / 0.278	0.052 / 0.166	0.094 / 0.095	0.171 / 0.292	0.130 / 0.118
Overcrowded dwelling	0.092 / 0.212	0.055 / 0.072	0.098 / 0.095	0.173 / 0.201	0.137 / 0.093
No car	0.061 / 0.199	0.032 / 0.103	0.111 / 0.064	0.057 / 0.388	0.159 / 0.078
No telephone or mobile phone	0.109 / 0.205	0.054 / 0.208	0.094 / 0.075	0.172 / 0.344	0.131 / 0.077
No colour TV	0.108 / 0.235	0.057 / 0.092	0.094 / 0.081	0.171 / 0.455	0.132 / 0.053
No computer	0.088 / 0.198	0.038 / 0.131	0.091 / 0.104	0.128 / 0.359	0.125 / 0.147
No washing machine	0.110 / 0.231	0.053 / 0.374	0.094 / 0.111	0.175 / 0.341	0.130 / 0.129

Notes: The figures refer to 2018, and illustrate the relationship between energy poverty and households' living conditions, based on own analysis of joint SILC-HBS microdata from Hungary. For each combination of deprivation item (in rows) and metric (in columns), the relevant figures denote the energy poverty rate among households unaffected by the relevant deprivation item (left figure) and those experiencing distress in the same category (right figure). The relevant SILC-based variables used to create binary indicators of housing distress are as follows: HH040 ("Leaking roof or damp wall"), HH081 ("No bath or shower"), HH091 ("No indoor flushing toilet"), HS160 ("Not enough light"), HS170 ("Noise from street or neighbours"), HS180 ("Pollution or grime"), HS190 ("Crime or violence in the area"), HS110 ("No car"), HS070 ("No telephone or mobile phone"), HS080 ("No colour TV"), HS090 ("No computer") and HS100 ("No washing machine"). The indicator for overcrowded dwellings was calculated on the basis of variable HH030 and the relevant Eurostat [methodology](#).

This highlights that selecting a particular measure of energy poverty is not an innocuous technical choice, but can fundamentally determine which segments of the population are subsequently identified as energy poor. Panel B shows that differences in energy poverty rates are more muted along age and gender lines, and suggests that gender-based or inter-generational perspectives may offer limited insights to better understand and effectively tackle energy poverty (see [Papadimitrou et al. \(2023\)](#) or [Bardazzi and Pazienza \(2023\)](#) for comparison and further references).

Importantly, the Hungarian microdata also enables the study of energy poverty in the context of households' living and housing conditions. Given the typical disconnect between households' expenditures (HBS component) and living conditions (SILC component) in terms of information source, the current analysis is the first to explore how different forms of energy poverty are related to and driven by particular dwelling and housing conditions. [Table 2](#) presents the results of regressing different poverty outcomes on categorical indicators of dwelling type, dwelling size and housing tenure. Similarly to [Table 1](#), the parameter estimates capture systematic conditional differences across categories (relative to outright homeowners living in mid-sized semi-detached houses). The comparison of different regressions reveals substantial differences in terms of both poverty concentration and the particular topology of the poverty rates. The estimated gap between various housing segments is highly changeable: the marginal difference ranges from −11.9 p.p. to +13.8 p.p. between house and apartment dwellers, from −0.3 p.p. to +29.3 p.p. between small and large dwellings, and from −0.9 p.p. to +23.8 p.p. between outright owners and tenants. On the whole, it appears that consensual measures of energy poverty are relatively invariant to differences in dwelling conditions (even though reduced-rate tenants and dwellers in small apartments face a significantly higher poverty risk), while expenditure-based measures are more sensitive to differences in housing characteristics. The *LowExpense* metric, in particular, displays a substantially higher concentration of energy poverty among apartment dwellers, tenants and households in small dwellings. These results suggest that the choice of measurement tool exerts a disproportionate influence not only on which socio-demographic segments of the population are identified as energy poor, but also on which dwelling conditions appear particularly vulnerable or conducive to energy-related deprivations ([Romero et al., 2018](#); [Faiella and Lavecchia, 2021](#); [Cong et al., 2022](#)).

While rarely considered in existing measurement initiatives ([Energy Poverty Advisory Hub, 2022a, 2023](#)), it is also useful to assess the role of inadequate housing conditions in driving energy poverty. Indeed, the concept of energy poverty as a specific form of material and social deprivation is well established in the literature and the consensual EU measures of energy poverty are based on the same survey information as the headline indicators of material and social deprivation and multi-dimensional poverty ([Bouzarovski and Thomson, 2018](#); [Pelz et al., 2018](#)).²⁵ Focusing on unconditional differences in energy poverty along several dimensions of housing-related deprivations using the SILC-based data component, the figures in [Table 3](#) present, separately for each dimension and deprivation status, the energy poverty rate associated with both non-deprived (left figure) and deprived households (right figure). As before, the table reveals substantial idiosyncratic variation but also some highly systemic features. First, consensual metrics (*Arrears*, *HomeWarm*) of energy poverty are very strongly correlated with inadequate housing conditions: the corresponding national poverty rates are, respectively, 2.7 and 3.5 times higher among households living in inadequate conditions relative to others. Similar patterns are observed in relation to the *LowExpense* metric, which suggests that absolutist indicators of energy poverty perform better in identifying distress among the most vulnerable. Second, expenditure-based metrics of the relative type are highly insensitive (and occasionally negatively related to) differences in households' living conditions, implying that high energy-related spending may equally indicate housing-related deprivations among low-income households and inefficient or conspicuous energy consumption among higher-income households, or the combination of these ([Romero et al., 2018](#); [Eisfeld and Seebauer, 2022](#)).

The importance of inadequate housing conditions for energy poverty measurement is also confirmed by the simultaneous analysis of households' socio-demographic background, housing arrangement and living conditions. The regression output presented in [Table A.1](#) of the [Appendix A](#) shows that extending the analysis to housing-related deprivations considerably increases the explanatory power of the model for all

²⁵ See the dedicated Eurostat website for the headline indicators of poverty and social exclusion in the EU [here](#).

metrics of energy poverty under consideration.²⁶ The results also show that housing-related deprivations, such as a leaking roof or the lack of sanitary facilities, may increase (or decrease) the incidence of energy poverty in an economically large and statistically significant manner. In particular, these figures indicate that consensual forms of energy poverty (such as *Arrears*, *HomeWarm*) are almost exclusively concentrated in the lowest income quintile or among households suffering from inadequate living conditions.

6. The longitudinal dimension

An additional distinctive feature of the Hungarian microdata is the reliance on a rotating sample of households, and the availability of a longitudinal data component as a result. With around half of all sampled households surveyed repeatedly in 2017 and 2018, it becomes possible to assess the persistence and joint behaviour of energy poverty measures over time, both at the individual and aggregate levels.

The analysis of the longitudinal sample ($n = 11\,135$) reveals that the level and composition of energy poverty is rather stable over time. At the aggregate level, energy poverty in Hungary remained steady between 2017 and 2018 (48.8% vs. 45.2%), with its dimensional structure and typical depth also remaining largely similar across households. Surprisingly, this is far from being the case at the micro level: almost half of repeatedly sampled households (47.6%) experienced a change in their energy poverty profiles along the five dimensions considered. Most transitions took place from a single-dimension energy poverty status to non-poverty status (13.8%), or the other way around (10.9%). Movements to or from multi-dimensional poverty states were less frequent (10% and 6.7%, respectively). Fig. 6 shows the relevant transition patterns, and reveals that much higher share of energy poor households have experienced change in their poverty status than the share of those that reported identical deprivation status in both years (71.5% vs. 28.5%). On the one hand, this suggests that a particular form of energy poverty is a less deep-seated, more transient form of social deprivation than other (more generic) forms of poverty. On the other hand, this also means that identifying vulnerable households as well as designing and identifying effective policies may be even more challenging than in other domains of social policy, given the required holistic perspective.

Zooming in on the transitions patterns among energy poor households can potentially reveal systemic movements between different forms of energy poverty and help identify the typical phases of energy-related deprivations among households. Specifically, using three-way step functions to characterise movements in and out of specific energy poverty dimensions (i.e. +1/0/−1 for in/stay/out), one may calculate the proximity of transitions in relation to each pairwise combination of measures.²⁷ The relevant proximity scores reveal that most transitions in or out of a particular form of energy poverty are statistically independent of transitions in other dimensions.²⁸ Real simultaneity is observed only in relation to the *HighShare* and *FixThreshold* metrics (with proximity scores of .267 and .228, respectively), but the majority of transitions take place in an asynchronous manner even

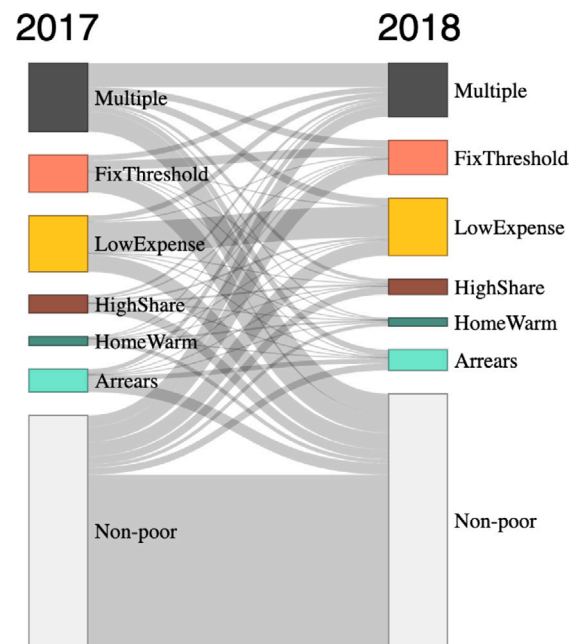


Fig. 6. Transition patterns in energy poverty across the years. Notes: Own illustration based on joint Hungarian HBS-SILC microdata from 2018, produced with the R packages 'networkD3' and 'tidyverse'. The nodes represent the relative share of population subject to energy poverty in zero (non-poor), singular (*Arrears*, *HomeWarm*, *HighShare*, *LowExpense*, *FixThreshold*) and multiple dimensions, respectively, in a given year. The links across the nodes indicate the observed transition patterns across nodes from 2017 to 2018.

in such relations. This randomness of transitions is consistent with the idea that financially-constrained (or energy-inefficient) households rationally self-select into different (and sometimes mutually exclusive) forms of energy poverty. Given that the relative costs and benefits of this choice are likely to be changeable both in the cross-section and over time, the degree of (in)adequacy of current measurement based on a single or limited subset of energy poverty metrics may also be equally variable.

7. Conclusions and policy recommendations

The empirical analysis presented in this study exploited the distinctive features of joint SILC-HBS national microdata from Hungary in order to study important and previously unexplored patterns of energy poverty in the context of EU measurement and policy. The main conclusion that emerges from the detailed comparison of existing measures of energy poverty is that different forms of energy deprivation concern largely different population segments. Households tend to self-select into different forms of energy poverty, and those that struggle to keep their homes warm or pay their utility bills are typically not the ones that spend uncharacteristically little or much on energy services. With less than one third of the energy poor suffering from multiple forms of energy-related deprivations, most forms of energy poverty tend to remain hidden under the existing measurement framework in the EU.

The empirical analysis also yields a series of additional novel insights:

- measures of energy poverty are subject to considerable seasonality that may seriously compromise the consistency and comparability of poverty outcomes;

²⁶ The relevant coefficients of determination (or R-squared statistics) range between .067–.305 in the extended specifications (Table A.1), compared to .027–.232 and .017–.132 when only information on households' socio-demographic background (Table 1) and dwelling characteristics (Table 2), respectively, is used on the right hand side.

²⁷ The proximity between two dimensions is perfect (with a proximity score of 1) if all transitions in or out of one dimension are identical to the transitions in the other dimension for all households. The proximity between two dimensions is minimal (with a proximity score of −1) if all transitions in one dimension are the opposite of the transitions observed in the other dimension.

²⁸ Most proximity scores for cross-dimension transitions are between −0.04 and 0.10, with a mean score of 0.06.

- the relationship between energy poverty and household income is highly variable across different indicators, but most forms of energy poverty are also present among non-financially constrained households;
- the socio-demographic profiles associated with various measures of energy poverty are markedly different; settlement type and household size appear to be the main drivers of poverty concentrations, while the role of age and gender is rather limited;
- living conditions play a highly important role in explaining horizontal differences in energy poverty; expenditure-based measures are more sensitive to differences in housing arrangements, while consensual measures are driven mainly by housing-related deprivations and inadequacies;
- while the rate of energy poverty appears robust at the aggregate level, longitudinal analysis shows limited persistence at the individual level: almost half of the energy poor change status or transit in or out of poverty from one year to the next.

These findings are based on the empirical analysis of Hungarian microdata. While energy poverty in Hungary exhibits strong path-dependencies typical of the CEE region as whole, it is reasonable to assume that broadly similar patterns apply equally to other EU Member States and advanced economies, as well. First, households' energy situation in Hungary is comparable to that of the typical EU country: the analysis of EU-wide microdata from 2015 shows that the mean energy poverty rate in Hungary is almost identical to the typical EU country across the five indicators considered (12.7% vs. 12.9%). Second, some of the presented findings – such as high seasonality or limited overlap between certain indicators – are structural and stem from existing data collection protocols and measurement choices. Third, previous studies from the United Kingdom and selected EU Member States have come to similar conclusions as regards the disconnect between existing indicators and their tendency for misspecification (Curtis and Pentecost, 2015; Heindl, 2015; Palmer et al., 2008; Romero et al., 2018).

More fundamentally, the presented analysis reveals serious shortcomings and limitations of all single-dimensional measurement of energy poverty. Consensual measures are subject to strong seasonality bias, exhibit limited cross-country comparability and display low correlations with observable aspects of households socio-economic background and living conditions. Expenditure-based measures, while more objective and robust, are unable to correctly identify households in need and not available in a timely and harmonised manner across all EU Member States (Herrero, 2017). A further potential problem with all the indirect measures in circulation is that they are hardly actionable for policy, as targeting support based on self-reported valuations or expenditure ratios is not feasible, equitable or sustainable.

This calls for immediate improvements to the current measurement framework, along the lines of an emerging consensus in the relevant literature. First, it advocates a shift towards well-designed multi-dimensional measurement initiatives that allow for the simultaneous consideration of a variety of vulnerability factors that influence the energy situation of households (such as energy efficiency, comfort, health, and well-being) and help eliminate most hidden forms of energy poverty (Herrero, 2017; Castaño-Rosa et al., 2019; Al Kez et al., 2024). This should also highlight the strong linkages of energy poverty with multi-dimensional poverty and deprivation under the international sustainable development agenda, and help advance its conceptualisation and recognition as a common and distinct research and policy concern across the Global North and South.²⁹ Second, it necessitates the further improvement of the EU survey data architecture and the harmonisation of European household surveys in particular. Related efforts should

focus on the synchronisation of data collection periods across EU Member States, the integration of HBS and SILC surveys towards common definitions and overlapping samples, as well as the revision, refinement and extension of specific survey questions for improved clarity and consistency. Third, the presented findings also call for exploring new avenues for the consistent measurement of previously omitted dimensions of energy poverty. The cornerstone of such initiatives should be the direct assessment of households' energy consumption and energy efficiency, on the basis of administrative sources and common minimum standards (Thomson et al., 2017). Subsequently, this would make it possible to calculate energy budgets or minimum income bundles based on minimum consumption thresholds and energy unit costs in a manner that is customised to the particular household segment and may incentivise energy savings and green investment (Romero et al., 2018; Menyhért et al., 2021).

As far as policy implications are concerned, the multi-faceted character of energy-related deprivations suggests that broader infrastructural, institutional and societal factors are at play (Bouzarovski et al., 2016). Therefore, while the measurement of energy poverty may focus only on its immediate drivers (i.e. household incomes, energy prices, and the energy efficiency of the dwelling), sound policy-making also requires a detailed understanding of the systemic pathways and conditions leading to social isolation and material deprivation in the home. Energy vulnerability and resilience are emerging as novel concepts that aim to bring different policy areas together to effectively combat energy poverty and insecurity in its diverse forms (Middlemiss and Gillard, 2015; Prime and Slabe-Erker, 2020). The particular Hungarian case under review, however, also reveals that different segments of the population tend to be energy deprived in different ways and more traditional and/or modular policy frameworks also stand a chance of effectively combating energy poverty. Specifically, the successful application of the standard toolkit of social and housing policy (e.g. targeted income support, home repair assistance, housing subsidies) may in itself eliminate a significant portion of energy poverty, especially among the most vulnerable (OECD, 2021; Menyhért, 2023). This also highlights that energy poverty and other forms of societal deprivations are intimately related, and the realisation of the EU2030 targets on poverty and social exclusion and the EU2030 targets on climate and energy are but two sides of the same coin.

CRediT authorship contribution statement

Bálint Menyhért: Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Bálint Menyhért reports financial support was provided by European Commission Joint Research Centre Ispra. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Appendix A

See Table A.1.

²⁹ The analysis of the SILC data component of the Hungarian microdata also shows that the energy poor population strongly overlaps with those at risk of poverty and social exclusion (AROPE) and monitor the EU 2030 target on poverty and social exclusion.

Table A.1

Extended regression of energy poverty on households' combined socio-demographic and housing background.

	Dependent variable				
	Arrears	HomeWarm	HighShare	LowExpense	FixThreshold
Income quintile #1	0.123 (.008)	0.066 (.006)	0.187 (.008)	0.227 (.009)	0.043 (.009)
Income quintile #2	0.037 (.008)	0.017 (.006)	0.037 (.007)	0.084 (.008)	0.021 (.008)
Income quintile #4	−0.019 (.007)	−0.011 (.006)	−0.030 (.007)	−0.041 (.008)	−0.024 (.008)
Income quintile #5	−0.035 (.007)	−0.001 (.006)	−0.051 (.007)	−0.053 (.008)	−0.066 (.008)
Urban area	−0.002 (.006)	0.002 (.005)	0.000 (.006)	0.040 (.007)	−0.034 (.007)
Rural area	0.024 (.006)	−0.007 (.005)	0.022 (.006)	0.005 (.007)	0.008 (.007)
Small household	−0.045 (.006)	0.014 (.004)	0.047 (.005)	0.085 (.006)	0.059 (.006)
Large household	0.044 (.008)	0.011 (.006)	−0.045 (.007)	−0.040 (.008)	−0.020 (.008)
Detached house	0.008 (.010)	0.027 (.007)	0.019 (.009)	−0.028 (.011)	0.066 (.011)
Apartment (small building)	0.017 (.015)	0.011 (.011)	−0.025 (.014)	0.076 (.016)	−0.025 (.016)
Apartment (large building)	0.015 (.011)	−0.012 (.008)	−0.020 (.010)	0.104 (.012)	−0.026 (.012)
Other accommodation	−0.007 (.022)	0.000 (.018)	0.156 (.022)	−0.055 (.025)	0.013 (.025)
1 or 2 rooms	−0.002 (.009)	−0.006 (.007)	0.001 (.008)	0.104 (.009)	0.019 (.010)
5 or more rooms	0.021 (.006)	0.009 (.005)	−0.021 (.006)	0.012 (.007)	−0.022 (.007)
Owner with mortgage	0.044 (.007)	0.009 (.005)	−0.015 (.006)	0.034 (.007)	0.006 (.007)
Tenant (market rate)	0.029 (.011)	0.003 (.009)	−0.039 (.011)	0.029 (.012)	−0.059 (.013)
Tenant (reduced rate)	0.060 (.014)	0.033 (.010)	−0.058 (.013)	0.108 (.015)	−0.053 (.015)
Free accommodation	0.007 (.010)	−0.024 (.008)	−0.013 (.010)	0.065 (.012)	−0.044 (.012)
Leaking roof or damp walls	0.126 (.006)	0.060 (.005)	−0.024 (.006)	−0.043 (.007)	−0.021 (.007)
No bath or shower	0.126 (.030)	0.028 (.023)	−0.055 (.029)	0.050 (.033)	−0.082 (.034)
No indoor flushing toilet	−0.083 (.029)	0.057 (.022)	−0.014 (.028)	−0.003 (.032)	0.046 (.032)
Not enough light	0.047 (.009)	0.020 (.007)	0.007 (.009)	0.020 (.010)	−0.003 (.010)
Noise from street or neighbours	−0.030 (.009)	−0.004 (.007)	−0.014 (.009)	−0.007 (.010)	0.000 (.010)
Pollution or grime	0.051 (.009)	0.030 (.007)	0.039 (.008)	0.001 (.010)	0.009 (.010)
Crime or violence in the area	0.039 (.012)	0.044 (.009)	−0.006 (.011)	0.031 (.012)	0.002 (.013)
Overcrowded dwelling	0.019 (.008)	−0.019 (.006)	−0.011 (.008)	−0.045 (.009)	−0.016 (.009)
No car	0.070 (.006)	0.017 (.004)	−0.102 (.006)	0.198 (.006)	−0.114 (.007)
No telephone or mobile home	−0.018 (.015)	0.018 (.012)	−0.039 (.015)	−0.013 (.017)	−0.045 (.017)
No colour TV	0.051 (.016)	−0.027 (.012)	−0.011 (.015)	0.166 (.018)	−0.036 (.018)
No computer	0.006 (.007)	0.019 (.005)	−0.025 (.007)	0.016 (.008)	0.022 (.008)
No washing machine	−0.062 (.022)	0.209 (.017)	0.021 (.021)	−0.099 (.024)	0.050 (.024)
Constant	−0.004 (.011)	−0.015 (.009)	0.097 (.011)	−0.012 (.012)	0.141 (.013)
R squared	.170	.105	.101	.305	.067
Nr of obs.	15 508	15 508	15 508	15 508	15 508

Notes: The presented results are based on sample-weighted OLS regressions featuring different metrics of energy poverty as the dependent variable and categorical variables on individuals' and households' socio-demographic background on the right hand side, using the full-cross section of the 2018 wave of the joint Hungarian SILC-HBS microdata. The reported estimates represent conditional mean differences in the energy poverty rate across different population segments as compared to the reference category – medium-sized households (3–4 members) in the 3rd equivalised income quintile living in moderately densely populated areas as outright owners in semi-detached homes of medium size (3–4 rooms) under adequate housing conditions – whose poverty rate is captured by the constant term. For the detailed explanation of housing-related deprivation items under consideration, see the notes to Table 3. Standard errors are in parenthesis, with statistical significance denoted by stars at the usual probability levels (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

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